

Nikolai Daelemans

nikolai@daelemans-experts.eu | <https://www.linkedin.com/in/nikolai-daelemans/> | www.nikolaidaelemans.com

SUMMARY

Software Engineer and AI Platform Engineer with 3+ years of experience building AI automation systems, cloud infrastructure, and full-stack applications. Currently leading infrastructure automation and AI platform engineering at KBC while co-founding AvantifAI, an enterprise AI consultancy. Proven track record delivering production AI systems (LLMs, RAG, agentic AI), IaC automation across 5+ systems, and scalable microservices in regulated environments. Open to engineering roles in AI/ML engineering, platform engineering, or infrastructure automation — hybrid or remote.

SKILLS

Languages:	Python, Java, C#, JavaScript, TypeScript, C++, SQL
AI / ML:	PyTorch, TensorFlow, LLMs, RAG, Vector Databases, NLP, Prompt Engineering, MLOps, ML Engineering, Agentic AI, GenAI
Frontend:	React, Angular, Vue, Flutter, HTML, CSS
Infrastructure:	Azure, AWS, Docker, Kubernetes, IaC, CI/CD, Kafka, Redis, Elasticsearch, Jenkins, Git
Databases:	PostgreSQL, SQL, Vector Databases
Data:	Power BI, Microsoft Fabric, Kibana, PySpark, Pandas, NumPy, Plotly
Networking:	Firewall, Zero Trust, IDS/IPS, NAS, SAN, GSLB, Netscaler, F5, Load Balancing (L4/L7), DDI/DNS
Methods:	Agile, Scrum, Microservices, REST API, DevOps, Product Ownership
Certs:	Microsoft Azure AI Fundamentals (2025); DataCamp: Efficient AI Model Training with PyTorch; Introduction to Power BI
Spoken Languages:	Dutch (C2), English (C2), French (B2), German (A2), Luxembourgish (A2)

PROFESSIONAL EXPERIENCE

CTO & Co-founder — AvantifAI (www.avantifai.com)

Based in Europe (March 2026 - present)

- Built the full MVP in under **4 weeks**: GDPR-compliant, multi-tenant SaaS with Dev/Prod environment separation, plan management, rate limiting, request throttling, and application-level security controls
- Architected the AI layer: **Voyage AI embeddings**, automated workspace vectorization pipeline, knowledge/brain graph for cross-document reference mapping, and a background job queue for async processing
- Auth0 for authentication, Sentry for monitoring, deployed on Railway (backend) and Vercel (frontend) with automated database backups
- Identified a gap in the **\$7B+** technical documentation market — teams lose hours per week to duplicated, stale, and structurally broken docs — and built **DocAI** to fix it; currently converting **13 warm leads** into **pilots**.
- DocAI scans entire documentation workspaces, surfaces duplication, stale content, and folder structure issues, and applies AI-driven fixes — reducing manual documentation overhead from hours to minutes

Infrastructure Software Engineer (AI Platforms & Automation) — KBC

Leuven, Belgium (July 2025 - present)

- Built reusable **IaC frameworks** automating provisioning across 5+ infrastructure systems including compute, storage, firewall, load balancing and DNS, with AI-assisted decision making.
- Automated infrastructure requests end-to-end: on-prem virtualization, VMware/SAN/NAS/XP8 storage, firewall rules (Zero Trust, IDS/IPS), L4/L7 load balancing, content switching, GSLB virtual servers, and DDI/DNS — using F5 and Netscaler.
- Team lead** of KBC's draining tool — organizing sprints and managing delivery in an agile environment using Jira.
- AI Innovation & Engineering** - Developing custom AI models and integrating Microsoft Copilot and Power Automate solutions to streamline internal banking processes. Troubleshooting recurring issues reported via internal support channels and automating resolutions to reduce human error.

Technology Consultant — PwC

Brussels, Belgium (February 2025 - May 2025)

- Led the organization and technical preparation together with Microsoft of a hackathon for the NATO, coordinating multi-disciplinary teams and aligning deliverables with defense innovation goals, resulting in actionable MARCOM data solutions and strategic prototypes that informed NATO maritime safety protocols.
- Automated a time-intensive multi-excel workflow used by cross-functional teams, developing a Python script to ingest, process, and insert records, eliminating manual data entry and saving 20+ engineering hours per month.
- Preprocessed and cleaned large-scale defense datasets using Python (Pandas, NumPy), enabling accurate analytics and feature engineering.
- Developed dynamic visualizations and dashboards with Power BI and Plotly to highlight key patterns and threats from complex data structures, improving interpretability and decision-making for NATO analysts and leadership.

Software Engineer — Tejo

Hybrid (October 2024 - December 2024)

- Designed and built a full event payment pipeline for a local non-profit — hosted frontend, Stripe/payment integration, automated confirmation emails with generated QR codes for event entry. Handled 100% digital payment processing with zero manual intervention on event day.

Software Engineer (Product Owner) — Belfius

Belgium (September 2024 - December 2024)

- Coordinated go-to-market technical readiness for an internal product modernization at Belfius, aligning engineering deliverables with business launch milestones and stakeholder requirements improving system performance by 50%.
- Designed and deployed scalable microservices using .NET Core and REST APIs, while contributing to Angular and React-based front-end enhancements to improve UX for internal users.

Lead IT & Analytics — Emergent

Leuven, Belgium (March 2024 - March 2025)

- Led the digital strategy and analytics initiatives for a student-run AI & Data Science organization, partnering with top-tier companies such as **McKinsey**, **KBC**, **AWS**, **Deloitte**, **DataCamp** and more to host workshops, talks, and hands-on projects; enhanced community engagement by 30% through scalable web solutions (WordPress/Elementor) and analytics-driven content planning (Python, PySpark).

Software Engineer (App Development) — *SunnySet*

Hybrid (January 2024 - September 2024)

- Built and deployed a cross-platform mobile app (Flutter, Java) delivering real-time safety alerts and local news for tourists.
- Led contributions across app logic, UI, and hosting; integrated CI/CD pipelines to ensure consistent performance across devices.

Software Engineer — *Think-E*

Leuven, Belgium (July 2023 - September 2023)

- Developed energy-efficient smart home applications by integrating IoT devices with real-time monitoring systems across residential complexes; leveraged C++, JavaScript, and SQL to connect sensors via Raspberry Pi, optimizing sustainability data collection and automation.
- Collaborated on full-stack solutions to improve energy tracking dashboards, incorporating responsive UI (HTML/MUI/CSS/JS) and GitHub-based version control to support scalable, maintainable systems.

EDUCATION

Bachelor of Science, Applied Computer Science June 2025

UCLL (University Colleges Leuven-Limburg), Belgium

PROJECTS

2025 — AI & Infrastructure Learning Platform (*In Progress*): Personal project building a learning platform for AI and infrastructure concepts. Hands-on experimentation with Raspberry Pi, IoT sensors, and custom model training. Focused on bridging theory and practical implementation.

2024 — AI Interview Analysis Platform: Full-stack interview platform built for a university AI course. Interviewers could manually create or AI-generated questionnaires; candidates completed these via a dedicated interface. Post-submission, interviewers received an analytics dashboard with candidate scoring, strength/weakness breakdowns, a personality matrix (e.g. innovation, teamwork), and full answer reviews. Built with Python, NLP, and LLMs.

2023 — Campus Assistant Chatbot (<https://github.com/NikolaiDoesCode/CampusAI>) Hybrid chatbot combining a custom-trained PyTorch intent classifier with DialoGPT fallback. Bag-of-words preprocessing pipeline, 5-layer MLP, deployed via Flask with a responsive frontend.

2022 — FaceDetectorCNN (<https://github.com/NikolaiDoesCode/Face-Detection>): Built a real-time face detection system from scratch using a custom CNN in PyTorch. Trained on 26,000 samples (LFW + generated negatives), achieving 99.89% validation accuracy and AUC-ROC of 1.0. Implemented sliding window detection, image pyramid, and non-maximum suppression from scratch.